Sinong (Simon) Zhan

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Education

Northwestern University Sep 2023 – Present
Ph.D. in Electrical and Computer Engineering **Expected Dec 2027**
Advisor: Prof. Qi Zhu

Northwestern University Sep 2023 – Jun 2026
Master's in Computer Engineering; advisor: Prof. Qi Zhu

University of California, Berkeley Aug 2018 – Dec 2022
B.A. in Computer Science and Applied Mathematics
Research advisor: Prof. Sanjit A. Seshia

Research Agenda

My research centers on **safe embodied agents**: how to translate human safety requirements into precise specifications, evaluate agent behavior, and train policies that act reliably in physical environments. I combine **formal methods, reinforcement learning, and foundation models** to connect safety evaluation with policy improvement. Robustness and efficiency are central to this agenda: agents must handle latency, uncertainty, and distribution shift while learning from limited data and computation.

- **Specification-grounded safety.** Temporal-logic specifications and runtime verification for evaluating foundation-model agents across interpretation, planning, and execution; safety-annotated data for improving robotic manipulation policies.
- **Certified and robust learning.** Barrier certificates and constrained policy optimization for safe control; belief-based learning for reliable decisions under observation delays and stochastic dynamics.
- **Efficient learning and adaptation.** Offline RL, sample-efficient reward learning, and memory-aware agent post-training to reduce interaction and context costs in long-horizon decision-making.

Research Experience

IDEAS Lab, Northwestern University Sep 2023 – Present
Graduate Research Assistant *Advisor: Prof. Qi Zhu*

- Co-developed **SENTINEL**, a formal safety evaluation framework for foundation-model embodied agents that checks semantic interpretation, action plans, and execution trajectories against temporal-logic specifications in VirtualHome and AI2-THOR (**NeurIPS 2026**).
- Co-developed **ManiGuard**, a specification-grounded manipulation benchmark and data suite with **200 tasks, 1,000 evaluation scenarios, and 8,000 safety-annotated demonstrations**; evaluates safety independently of task success under distribution shift and on a physical Franka platform.
- Developed **delay-robust RL** using variational policy optimization, direct belief forecasting, and transformer-based offline belief-policy learning; publications at NeurIPS 2024 (**Spotlight**), ICML 2024/2025, and ICLR 2026.
- Developed **state-wise safe RL from pixels** by coupling visual world models with probabilistic barrier-certificate learning (L4DC 2024), and transition-aware reward shaping for robust, sample-efficient inverse RL in stochastic environments (L4DC 2026).

University of California, Berkeley Feb 2022 – Apr 2023

Undergraduate Researcher & Research Assistant *Advisor: Prof. Sanjit A. Seshia*

- Developed automated verification and control-synthesis methods for robotic systems, including path planning and trajectory optimization for unmanned underwater vehicles.
- Built an SMT-based 3D bin-packing solver for design model checking and automated verification workflows for the NSF UUV Challenge.

Human Computing Lab, ISCAS & XDiscovery Lab, Dartmouth May 2019 – Sep 2021

Research Assistant

Advisors: Prof. Feng Tian, Prof. Teng Han, and Prof. Xing-Dong Yang

- Developed microfluidics-based sensing technologies and VR interaction models with real-time feedback, leading to CHI 2021 and UbiComp 2022 publications.
- Implemented self-supervised methods for multimodal sensor fusion in haptic interfaces and explored rapid prototyping for human-computer interaction.

Industry Experience

Google Jun 2026 – Present

Research Intern *Google Research & P+D Team*

- Built end-to-end RL training infrastructure for long-horizon coding agents on Google XManager, supporting post-training of **0.5B–30B parameter models** with tensor and sequence parallelism, ZeRO-1 optimizer sharding, and memory management for colocated rollout and training.
- Designed summarization-based RL that learns to retain informative trajectory context while interleaving reasoning and tool use; developed separate credit assignment for summarization, reasoning, and tool-calls.

Amazon Jun 2025 – Sep 2025

Applied Scientist Intern *Store Foundation AI Team*

- Developed multi-turn conversational agents for the Rufus Shopping Assistant using RL, and designed logic-based reward machines for human behavior simulation.
- Optimized LLM-agent training workflows and contributed to customer-facing systems using AWS infrastructure, including EC2/EKS, S3, and CloudWatch.

Publications

*Papers and project links: simon-zhan.com/publications. * indicates equal contribution.*

Conference and Journal Papers

1. **S. S. Zhan***, P. Wang*, J. Liu*, Y. Peng*, Y. Lyu, Z. Wang, Q. Wang, Z. Ruan, X. Shi, X. Cao, F. Yang, Z. Ni, K. Wang, R. Zhang, H. Shao, M. Li, Q. Zhu. "SENTINEL: A Multi-Level Formal Framework for Safety Evaluation of Foundation Model-based Embodied Agents." *NeurIPS, 2026 (accepted).*
2. W. Feng, L. Wang, T. Wei, J. Zhang, C. Gao, **S. S. Zhan**, P. Lv, W. Dong. "Token Buncher: Shielding LLMs from Harmful Reinforcement Learning Fine-Tuning." *CCS, 2026 (accepted).*
3. C. Li, W. Tang, Y. Huang, **S. S. Zhan**, M. Hu, X. Jia, Y. Liu. "Shedding Light on VLN Robustness: A Black-box Framework for Indoor Lighting-based Adversarial Attack." *CVPR, 2026.*
4. X. Shi, J. Ding, X. Zhao, **S. Zhan**, P. Mohapatra, D. Quispe, K. Welbeck, J. Cao, W. Chen, P. Guo, Q. Zhu. "STEP-LLM: Generating CAD STEP Models from Natural Language with Large Language Models." *DATE, 2026.*

5. **S. S. Zhan***, P. Wang*, Q. Wu*, R. Jiao, Y. Wang, C. Huang, Q. Zhu. "Enhancing Inverse Reinforcement Learning through Encoding Dynamic Information in Reward Shaping." *L4DC, 2026.*
6. **S. S. Zhan***, Q. Wu*, P. Wang, F. Yang, X. Shi, C. Huang, Q. Zhu. "Belief-Based Offline Reinforcement Learning for Delay-Robust Policy Optimization." *ICLR, 2026.*
7. Y. Zhang, T. Wang, J. Gesi, Z. Wang, Y. Lu, J. Lin, **S. Zhan**, V. Gao, R. Jiao, J. Liu, K. Qian, Y. Tang, R. Xue, H. Zhang, Q. Cui, Y. Guo, D. Wang. "Shop-R1: Rewarding LLMs to Simulate Human Behavior in Online Shopping via Reinforcement Learning." *ICLR, 2026.*
8. Q. Wu*, Y. Wang*, **S. S. Zhan***, Y. Wang, C.-W. Lin, C. Lv, Q. Zhu, J. Schmidhuber, C. Huang. "Directly Forecasting Belief for Reinforcement Learning with Delays." *ICML, 2025.*
9. Q. Wu*, **S. S. Zhan***, Y. Wang, Y. Wang, C.-W. Lin, C. Lv, Q. Zhu, C. Huang. "Variational Delayed Policy Optimization." *NeurIPS, 2024.* ***Spotlight.***
10. Q. Wu, **S. S. Zhan**, Y. Wang, Y. Wang, C.-W. Lin, C. Lv, Q. Zhu, J. Schmidhuber, C. Huang. "Boosting Reinforcement Learning with Strongly Delayed Feedback Through Auxiliary Short Delays." *ICML, 2024.*
11. **S. S. Zhan**, Y. Wang, Q. Wu, R. Jiao, C. Huang, Q. Zhu. "State-wise Safe Reinforcement Learning With Pixel Observations." *L4DC, 2024.*
12. R. Jiao*, Y. Wang*, X. Liu, **S. S. Zhan**, C. Huang, Q. Zhu. "Kinematics-aware Trajectory Generation and Prediction with Latent SDE." *IROS, 2024.*
13. F. Yang, **S. S. Zhan**, Y. Wang, C. Huang, Q. Zhu. "Case Study: Runtime Safety Verification of Neural Network Controlled System." *RV, 2024.*
14. H. Su, S. Feng, **S. S. Zhan**, N. Zhan. "Switching Controller Synthesis for Hybrid Systems Against STL Formulas." *FM, 2024.*
15. Y. Wang, **S. S. Zhan**, R. Jiao, Z. Wang, W. Jin, Z. Yang, Z. Wang, C. Huang, Q. Zhu. "Enforcing Hard Constraints with Soft Barriers: Safe Reinforcement Learning in Unknown Stochastic Environments." *ICML, 2023.*
16. Y. Wang*, **S. S. Zhan***, Z. Wang, C. Huang, Z. Wang, Z. Yang, Q. Zhu. "Joint Differentiable Optimization and Verification for Certified Reinforcement Learning." *ICCPS, 2023.*
17. W. Sun, Y. Chen, Y. Chen, X. Zhang, **S. Zhan**, Y. Li, J. Wu, T. Han, H. Mi, J. Wang, F. Tian, X.-D. Yang. "MicroFluID – A Reconfigurable RFID Platform for Robust Interaction Sensing Based on Microfluidics." *IMWUT / UbiComp, 2022.*
18. W. Sun, Y. Chen, **S. Zhan**, T. Han, F. Tian, H. Wang, X.-D. Yang. "RElectrode: A Reconfigurable Electrode For Multi-Purpose Sensing Based on Microfluidics." *CHI, 2021.*

Workshop Papers

19. Y. Zhang, J. Gesi, R. Xue, T. Wang, Z. Wang, Y. Lu, **S. Zhan**, H. Zeng, Q. Cui, Y. Guo, J. Huang, M. Shah, D. Wang. "See, Think, Act: Online Shopper Behavior Simulation with VLM Agents." *SEA Workshop at NeurIPS, 2025.*
20. Y. Wang, R. Jiao, **S. Zhan**, C. Lang, C. Huang, Z. Wang, Z. Yang, Q. Zhu. "Empowering Autonomous Driving with Large Language Models: A Safety Perspective." *LLMAgent Workshop at ICLR, 2024.*

Preprints

21. Y. Peng*, P. Wang*, **S. S. Zhan***, Y. Lyu, Z. Ni, J. Yan, F. Wong, R. Jiao, H. Yin, X. Cao, H. Shao, M. Li, R. Zhang, Q. Zhu. "ManiGuard: A Benchmark and Data Suite for Specification-Grounded Safety Evaluation and Improvement of Robotic Manipulation." *arXiv:2608.17386, 2026.*

22. Q. Wu, Y. Wang, **S. S. Zhan**, Y. Dai, S. Deng, S. Habchi, Q. Zhu, M. Gallé, C. Huang. "A Unified Framework for Rethinking Policy Divergence Measures in GRPO." *arXiv:2602.05494, 2026.*
23. **S. S. Zhan***, Q. Wu*, Z. Ruan, F. Yang, P. Wang, Y. Wang, R. Jiao, C. Huang, Q. Zhu. "Inverse Delayed Reinforcement Learning." *arXiv:2412.02931, 2024.*

Manuscript Under Review

Q. Wu, Y. Wang, **S. S. Zhan**, Q. Zhu, J. Schmidhuber, C. Huang. "Efficient Multi-Step Reinforcement Learning with Expectation-Maximization Bootstrapping."

Teaching and Mentoring

University of California, Berkeley Fall 2022
Teaching Assistant *Math 128A: Numerical Analysis*

- Led weekly discussion sections and office hours for **40+ students**; graded assignments and provided feedback on numerical methods and computational mathematics.

Research mentoring

- **Master's students:** Tianhang Zhu (2026–present); Yiyan Peng (2026–present); Zinan Wang (2025–2026; now at Moore Threads); Xinyu Cao (2025–2026).
- **Undergraduates:** Fiorelli Wong (2026–present); Jixin Yan (2026–present); Eric Yang (2025–present); Justin Liu (USC, 2024–2026; now Ph.D. at UC Berkeley EECS); Philip Wang (2024–2026; now Ph.D. at Northwestern ECE); Aria Ruan (2023–2025; now at Tesla); Frank Yang (2023–2025; now at Scale AI).

Academic Service

Conference reviewer: NeurIPS, ICML, ICLR, AAAI, CVPR, ECCV, ICRA, L4DC, ASP-DAC, ECC.
Journal reviewer: Machine Learning (Springer); IEEE Internet of Things Journal.
Committee: ICCPS Artifact Evaluation Committee.

Patents

1. **EMG-based Multi-target Control System.** Simon Zhan, Junjun Fan, Feng Tian, Wei Sun. Chinese patent/application no. CN201710953534.X.
2. **Microfluidic Pipeline Composite Structure and Pattern Deformation System.** Wei Sun, Yanjun Chen, Simon Zhan, Teng Han, Feng Tian, Hongan Wang, Xing-Dong Yang. Chinese patent/application no. CN202110377915.4.
3. **Fluid Pattern Reconfigurable System Based on Microfluidic Technology.** Wei Sun, Yanjun Chen, Simon Zhan, Teng Han, Feng Tian, Hongan Wang, Xing-Dong Yang. Chinese patent/application no. CN202110378536.7.

Technical Skills

Programming: Python, C/C++, CUDA, Java, R, Julia, SQL, MATLAB, Lean 4.
ML and agent training: PyTorch, JAX, Hugging Face, verl, TensorFlow.
Optimization and simulation: Gurobi, CVX, CasADi, MuJoCo, Isaac Sim, VirtualHome, AI2-THOR.
Systems: Linux, Git, Docker, AWS, Google XManager, Unity3D, Simulink, LaTeX.
Languages: English (native), Chinese (native), German (limited proficiency).